WEEK 5 – DATABASE USE IN LIBRARIES AND INFORMATION CENTRES

INTRODUCTION

The rapid growth of information and the increasing reliance on digital technologies have made databases central to modern library and information services. In contemporary libraries and information centres, information is no longer stored only in printed form. Instead, large volumes of data and documents are organised, stored, accessed, and retrieved through electronic databases.

Databases enable libraries to manage vast collections efficiently, provide timely access to information, support research activities, and meet the diverse information needs of users. A proper understanding of database use is therefore essential for library and information science students, as librarians are expected to manage, search, evaluate, and teach the effective use of databases in both academic and non-academic environments.

MEANING OF A DATABASE

A database can be described as a **systematic and organised collection of related data stored electronically** and designed for easy access, retrieval, updating, and management. Unlike random collections of files, a database is structured in a way that allows users to locate specific information quickly and accurately.

In libraries and information centres, databases may contain bibliographic records, full-text documents, multimedia resources, user records, circulation data, or research outputs. The organisation of data within a database ensures consistency, reduces duplication, and enhances information retrieval.

DATABASES IN THE CONTEXT OF LIBRARIES AND INFORMATION CENTRES

In the library environment, databases serve as the backbone of electronic information services. They support both internal library operations and external user services. Internally, databases manage cataloguing records, acquisitions data, circulation transactions, and user information. Externally, they provide users with access to scholarly literature, reference materials, statistical data, and digital content.

Libraries no longer rely solely on locally held collections. Through databases, they connect users to global information resources, thereby expanding the scope and relevance of library services.

TYPES OF DATABASES USED IN LIBRARIES AND INFORMATION CENTRES

Bibliographic Databases

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Bibliographic databases contain descriptive information about documents rather than the full text. They include details such as author, title, subject, publication date, and abstracts. These databases guide users to relevant sources and indicate where full texts can be obtained.

In academic libraries, bibliographic databases are widely used for literature searching and research planning.

Full-Text Databases

Full-text databases provide complete access to documents, including journal articles, books, conference papers, reports, and theses. Users can read, download, and sometimes print materials directly from these databases.

Full-text databases have significantly reduced the time and effort required to access scholarly information and have transformed academic research and learning.

Numerical and Statistical Databases

These databases store quantitative data such as statistics, survey results, economic indicators, and scientific measurements. They are commonly used in research, policy formulation, and data analysis.

Libraries support access to such databases to meet the needs of researchers in the social sciences, sciences, and humanities.

Multimedia Databases

Multimedia databases contain non-textual materials such as images, audio recordings, videos, maps, and animations. They support teaching, learning, and research across various disciplines.

Institutional Databases and Repositories

Institutional databases store locally generated information such as theses, dissertations, research articles, staff publications, and institutional reports. These databases promote visibility, preservation, and access to institutional knowledge.

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DATABASE MANAGEMENT SYSTEM (DBMS)

A Database Management System is the software used to create, store, manage, and retrieve data within a database. It acts as an interface between the user and the database.

In libraries, DBMS ensures data security, controls access, allows multiple users to work simultaneously, and supports data backup and recovery. Examples include MySQL, PostgreSQL, and database modules within library management systems.

DATABASE USE BY LIBRARY USERS

Library users interact with databases for various purposes. Students use databases to find academic materials for assignments and examinations. Researchers rely on databases for literature review, data analysis, and scholarly communication. Professionals use databases for decision-making, policy formulation, and continuing education.

Effective database use requires users to understand search strategies, database interfaces, and evaluation techniques. Libraries therefore provide training and user education programmes to enhance database literacy.

ROLE OF LIBRARIANS IN DATABASE USE

Librarians play a critical role in database selection, management, and use. They evaluate databases before subscription, ensuring relevance, quality, and cost-effectiveness. Librarians also organise training sessions, create user guides, and provide one-on-one assistance to users.

In addition, librarians monitor database usage statistics, troubleshoot access issues, and liaise with database vendors. Their expertise ensures that databases are fully utilised and aligned with institutional goals.

SEARCHING DATABASES IN LIBRARIES AND INFORMATION CENTRES

Database searching is more structured than general internet searching. It involves understanding controlled vocabularies, subject headings, indexing terms, and advanced search features.

Users can refine searches using search fields, filters, limiters, and logical operators. Mastery of these features improves retrieval precision and reduces irrelevant results.

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ADVANTAGES OF DATABASE USE IN LIBRARIES

Database use offers numerous benefits, including faster access to information, improved accuracy, remote availability, efficient storage, and enhanced research productivity. Databases also support resource sharing, collaboration, and digital preservation.

Through databases, libraries can serve more users with fewer physical constraints.

CHALLENGES OF DATABASE USE IN LIBRARIES AND INFORMATION CENTRES

Despite their benefits, databases present challenges such as high subscription costs, licensing restrictions, technical issues, poor internet connectivity, and limited user skills. In developing countries, inadequate infrastructure further complicates database access and use.

Addressing these challenges requires institutional support, continuous training, and strategic planning.

FUTURE OF DATABASE USE IN LIBRARIES

The future of database use in libraries is closely linked to advancements in artificial intelligence, big data, and cloud computing. Intelligent search systems, personalised recommendations, and enhanced analytics will further transform how users interact with databases.

Libraries must adapt to these changes to remain relevant in the digital information landscape.

CONCLUSION

Database use has become indispensable in libraries and information centres. Databases support efficient information organisation, retrieval, and dissemination, thereby enhancing the quality of library services. For librarians and information professionals, competence in database use is no longer optional but a core professional requirement.

A library that effectively manages and promotes database use contributes significantly to teaching, research, and national development.

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